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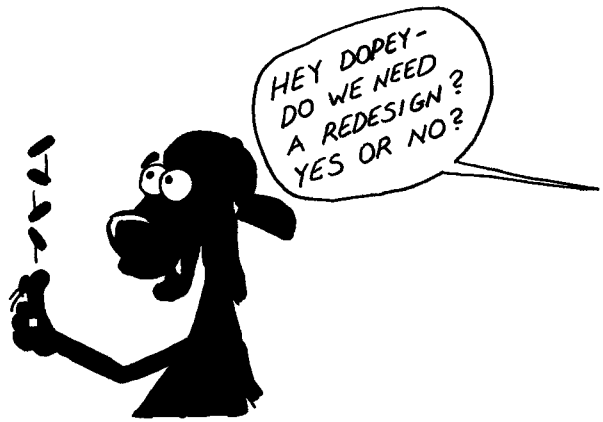
Seat-of-the-Pants Engineering



7.1 INTRODUCTION

The next few pages will examine explicitly what we've already seen in some of the homework problems—the question of how to get a *good enough* answer instead of an exact one. Although this chapter and its problems could have been broken up and distributed among the earlier chapters, the approach seems to be distinctive enough to merit a more unified presentation. To a large extent the ability to do good seat-of-the-pants engineering is an acquired skill, one that comes after having gone through the tedium of producing exact results for a great many problems. The difficulty for the student is that the whole idea of seat-of-the-pants analysis is very seldom referred to in any text even though it underlies much of the successful engineering of this world. Really good engineers rarely sit down and precisely analyze a problem at the start. Rather, they consider many options, get quick approximations, redesign, get some more approximate solutions, and then finally zero in on a final solution. Only after a particular problem has been adequately pinned down do really serious analysis tools come into play. This chapter attempts to give the student some sense of what can be accomplished by stepping back and thinking about a particular problem in a general way before bringing out all the analytical tools the preceding chapters have given us.

7.2 GETTING APPROXIMATE RESULTS

Seat-of-the-pants engineering refers to the ability to analyze a system reasonably well, *but not exactly*, in a short time. Obviously, you can analyze a given problem in great depth and detail if you're given the necessary time and resources. Nevertheless, it's often the case that you'll have to produce an answer without having all the time you might like. In addition to situations like these, you will sometimes want to generate an approximate answer quickly because you're considering different design options and you don't want to spend too much development time on an idea that's not going to pan out. Thus, there's a good deal of value in having the ability to get a reasonably good answer fairly quickly.

As an example of how we can cut quickly to an answer, let's revisit our spring-mass problem of Section 1.2. You'll recall that in this problem we were trying to model the system as a lumped spring and mass and to determine the combined natural frequency of the system. What made the analysis approximate was the fact that the spring element had mass of its own, mass that wasn't accounted for in the analysis. To illustrate how we can reasonably account for this added mass, we'll consider longitudinal motions of a bar with a lumped mass at one end, as shown in Figure 7.1.

If the bar actually was a pure spring, then the deflected position of the bar under an applied load at the end would vary linearly, from zero at the fixed end to a maximum value at the free end. In Section 1.2 we found the relation between the deflection at the end and the applied force (1.2.47) to be

$$\xi(l) = \frac{fl}{EA} \tag{7.2.1}$$

If our simplifying assumptions hold, then a linearly varying displacement along the bar implies that the velocities will vary linearly also, since finding the velocities simply involves differentiation in time. No spatial operations are involved, so the spatial relations stay the same.

We've already seen that the natural frequencies of a system can be found by taking ratios of the system energies (Rayleigh's quotient). Since we know what the velocity field looks like within the bar (if it behaves according to our assumptions), we can modify the kinetic energy part of Rayleigh's quotient and solve for the modified natural frequency. Originally, the kinetic energy was given by

$$KE = \frac{1}{2} m_1 \dot{\xi}^2(l) \tag{7.2.2}$$

That is, the only component of the kinetic energy was that due to the lumped mass at the right end. Now we're going to add the energy of the bar. If the velocity distribution were truly linear, we'd have

$$\dot{\xi}(x) = \frac{x}{l}\dot{\xi}(l) \tag{7.2.3}$$

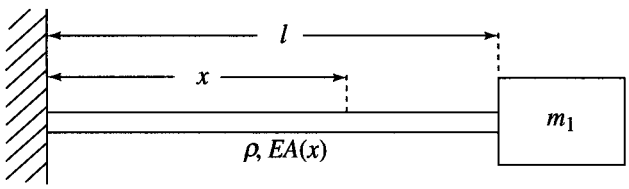


Figure 7.1 Uniform bar with a lumped mass at the right end

In reality, the actual distribution *won't* be precisely linear. For large end masses, the first mode will be very close to a linear distribution, and the closeness will degrade as the end mass is reduced with respect to the bar's mass.

Taking the bar's mass into account, our new kinetic energy becomes

$$KE' = \frac{1}{2}m_1\dot{\xi}^2(l) + \frac{1}{2}\int_0^l \rho\dot{\xi}^2(x)dx \tag{7.2.4}$$

or, using (7.2.3)

$$KE' = \frac{1}{2}m_1\dot{\xi}^2(l) + \frac{1}{2}\int_0^l \rho\left(\frac{x}{l}\right)^2\dot{\xi}^2(l)dx \tag{7.2.5}$$

We can factor out a couple of terms to leave us with

$$KE' = \frac{1}{2}\dot{\xi}^2(l)\left(m_1 + \int_0^l \rho\left(\frac{x}{l}\right)^2 dx\right) \tag{7.2.6}$$

If we now evaluate the integral, we'll be left with

$$KE' = \frac{1}{2}\left(m_1 + \frac{\rho l}{3}\right)\dot{\xi}^2(l) \tag{7.2.7}$$

Comparing (7.2.2) and (7.2.7), we see that what we've got is an equivalent mass. Whereas formerly our mass was just equal to m_1, now it's equal to $m_1 + \frac{\rho l}{3}$. Thus we can go ahead and use our old $\omega_n = \sqrt{\frac{k}{m}}$ formulation (with $k = \frac{EA}{l}$), but this time we'll use the modified mass rather than just the lumped mass. Therefore

$$\omega_n = \sqrt{\frac{EA}{l\left(m_1 + \frac{\rho l}{3}\right)}} \tag{7.2.8}$$

If we calculate the natural frequency for the case of $EA = 10$ N, $l = 1$ m, $m_1 = 10$ kg, and $\rho = 1$ kg/m, we'll find that $\omega_n = .9837$ rad/s. Without the correction for ρ, our estimate for the natural frequency is 1.0 rad/s, giving us an error of about 2% from the exact result of .9836 rad/s (found from solving the actual partial differential equation). By modifying our mass term, we've reduced the error in the estimate to about one-hundredth of a percent.

Let's next go ahead and tackle the tougher problem of $\rho = 10$ kg/m. For this case we previously (Section 1.2) had a 16% error, reflecting the fact that since the bar's mass was equal to the lumped mass, the assumption of a massless spring element in place of the bar became less reasonable. Our modified estimate for the natural frequency (using the new modified mass) is .8660 rad/s. If we compare this with the exact value of .8603 rad/s, we see that we've cut the error from 16% to about .5%.

These are major improvements! Just by using a little insight, and realizing that the motions of the bar will contribute to the system's kinetic energy, we've improved our frequency estimates tremendously. Of course, if you really want the exact solution, you can always use the techniques

of Chapter 5 to find it. But what we've gotten instead is a very accurate estimation with very little effort.

Okay, we've now seen how we can improve on our frequency estimate for an isolated SDOF problem. Let's move beyond that and into a MDOF estimation problem. For this example, we'll assume that you work for a major automobile manufacturer and your job involves suspension design. A colleague has come up with a very nice model of a vehicle, one that includes seven degrees of freedom. The precise model isn't important because we won't be doing an exact analysis, but Figure 7.2 shows the basic setup. The tires are modeled as spring elements, as are the suspension pieces between the axles and the body. The front and rear axles are included as rigid bars, and the body mass is supported by the suspension springs. Since the model has seven degrees of freedom, it also has seven eigenvectors, eigenvectors that can be associated with specific types of responses. For instance, the lowest natural frequency mode is associated with what automotive designers call *bounce.* This is the same mode that airplane designers call *heave*, and it corresponds to purely up-and-down motions of the body. Although the physical mode that's called bounce doesn't *exactly* correspond to pure up-and-down motions (there's some small amount of pitch, etc.), the approximation found by ignoring everything but the up-and-down motions is actually quite good. There's also a *roll* mode, which corresponds to rotations about the longitudinal axis of the car. *Pitching* refers to rotations about a lateral axis through the car (the sort of motion you'd get from strongly applying the brakes). These low frequency modes are *rigid body modes* because they reflect motions of the car body itself, oscillating on its suspension. The remaining modes are higher frequency modes that involve motions of smaller subsystems within the vehicle.

The data to be used in this example are taken from an actual car analysis discussed elsewhere [7]. In this model, there are two kinds of mass element. The car body is called a *sprung* mass because it's supported by the suspension. The mass elements that aren't directly attached to the body, like the rear axle, are called *unsprung* mass elements. Because these masses aren't directly connected to the body, they can experience their own, higher frequency modes. One of these is called *wheel hop,* which occurs when the front or back wheels bounce up and down in unison. If a set of wheels is moving in

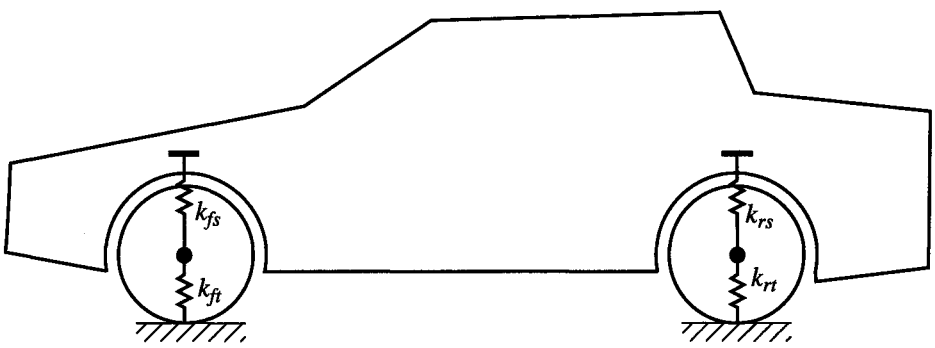


Figure 7.2 Simplified automobile vibration model

TABLE 7.1 PARAMETER VALUES FOR 1975 COMPACT TWO-DOOR

Parameters	Parametric values
Tire stiffness	184 kN/m
Front suspension stiffness	18.7 kN/m
Rear suspension stiffness	26.1 kN/m
Vehicle mass	1065 kg
Total unsprung mass	175 kg
Front unsprung mass	73 kg
Rear unsprung mass	102 kg
Rear axle	52 kg
Body roll inertia	190 kg·m^2
Body pitch inertia	1080 kg·m^2
Rear axle rotational inertia	8 kg·m^2
Wheelbase	2.42 m
Front wheels to CG	1.2 m
Rear wheels to CG	1.22 m
Ground to CG	.46 m
Front track	1.38 m
Rear track	1.35 m

an antisymmetric way (one up and the other down), the mode is called *tramp*. The various physical parameters of the vehicle model are given in Table 7.1.

To analyze the system and to determine what the frequencies and mode shapes are, you could construct the equations of motion for the complete system, code it up, and analyze it. But what we're going to show now is how you can get a quick and reasonable estimate of the responses without going to all this trouble. Just for comparison, Table 7.2 lists the exactly calculated natural frequencies for the modes we'll be looking at.

Let's start with the bounce mode. We know that this mode ideally consists of the body moving up and down. If you ignore the mass between the tire springs and the suspension springs, you can model them as two springs in series. Thus the equivalent spring at each front corner of the car is

TABLE 7.2 NATURAL FREQUENCIES FOR SOME CAR VIBRATION MODES

Vibration mode	Frequency of oscillation (Hz)
Bounce (sprung mass)	1.33
Pitch (sprung mass)	1.71
Roll (sprung mass)	2.27
Front-wheel hop	11.87
Front-tramp	11.87
Rear-wheel hop	10.19
Rear-wheel tramp	12.63

equal to

$$k'_f = \frac{k_{ft}k_{fs}}{k_{ft} + k_{fs}} = 16{,}975 \text{ N/m} \tag{7.2.9}$$

and

$$k'_r = \frac{k_{rt}k_{rs}}{k_{rt} + k_{rs}} = 22{,}858 \text{ N/m} \tag{7.2.10}$$

The subscripts f and r indicate whether we're dealing with the front or rear springs and the subscripts t and s indicate whether we're looking at the tire or the suspension spring. Thus k_{ft} is referring to the stiffness due to the tire at the front of the car.

Remember that we're after a single mode, the one corresponding to bounce. If we call x_b the coordinate of the bounce mode, then we can write

$$1065\ddot{x}_b + 2(16{,}975 + 22{,}858)x_b = 0 \tag{7.2.11}$$

Note that 1065 is the total mass (in kilograms) and 2(16,975+22,858) is the total spring constant. (We've got spring forces from both the left and the right, thus giving us the factor of 2.)

Solving for the natural frequency gives us

$$\omega_n = \sqrt{\frac{79{,}666}{1065}} = 8.65 \text{ rad/s} = 1.38 \text{ Hz} \tag{7.2.12}$$

This can be compared to the exact value found in Table 7.2, which is 1.33 Hz. Thus our quick-and-dirty solution gave us an answer that is within 4% of the actual answer.

Given this success, let's go crazy and try to replicate another answer. The exact solution for the roll mode gives a natural frequency of 2.27 rad/s for the sprung mass. The moment of inertia for the sprung mass about the longitudinal axis is 190 kg·m^2. All we need do is determine the angular stiffness. Figure 7.3 shows what we've actually got to deal with. When the car rolls, the equivalent springs at the front and back are deformed. As Figure 7.3 shows, the equivalent torsional spring for each spring is equal to $k_{\text{eq}}l^2$, where l is the distance from the roll center to the spring and k_{eq} corresponds to the k'_f and k'_r defined. Note that the distances from the roll center to the equivalent spring is given by the front and rear track data in Table 7.1.

Our equation of motion for the roll mode is therefore given by

$$190\ddot{x}_r + 2\left(16{,}975\left(\frac{1.38}{2}\right)^2 + 22{,}858\left(\frac{1.35}{2}\right)^2\right)x_r = 0 \tag{7.2.13}$$

For this case, the natural frequency turns out to be

$$\omega_n = \sqrt{\frac{34{,}598}{190}} = 14.0 \text{ rad/s} = 2.22 \text{ Hz} \tag{7.2.14}$$

a result that's off only by 2% from the exact solution.

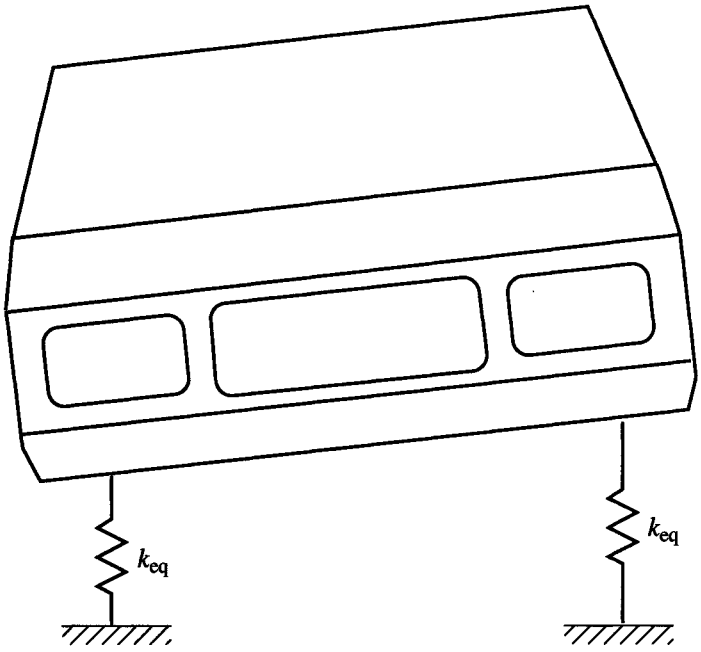


Figure 7.3 Roll mode

Finally, let's look at one more mode: that of front-wheel hop. This is an interesting one because we can choose from among a few different approximation levels. The most basic observation to make is that we've got a relatively large mass (890 kg) connected to a relatively small one (73 kg) by way of two suspension springs, each with stiffness equal to 18,700 N/m. The 890 kg mass corresponds to the sprung mass while the 73 kg mass represents the front unsprung mass (the mass between the tire spring and the suspension). The unsprung mass is sprung to ground by means of the two tire springs, each having stiffness equal to 184,000 N/m. If we completely ignore the sprung mass and suspension springs, we're left with the unsprung mass supported by two tire springs

$$73\ddot{x}_h + 2(184{,}000)x_h = 0 \tag{7.2.15}$$

This gives us a natural frequency estimate of 11.3 Hz., which is only 5% off from the exact value of 11.87 Hz. But wait, there's more! We observe that this frequency is much higher than the bounce frequency. Now, recall what happened in your physical experiment with the mass and spring when you moved your hand at different speeds. When you vibrated your hand quickly, the mass didn't move; it was isolated by means of the spring. Well, just think about our car. According to our estimates, if the rest of the car isn't moving, then the front-tire hop occurs at a frequency of 11.3 Hz. Let's make believe that the tires *are* bouncing at this frequency. Will the large mass of the car itself respond? No way! The total sprung vehicle mass is equal to 890 kg. The springs between this mass and the wheels have a combined spring constant of $2(18{,}700) = 37{,}400$ N/m. Thus the natural frequency of this

mass on these springs is

$$\omega_n = \sqrt{\frac{37{,}400}{890}} = 6.48 \text{ rad/s} = 1.03 \text{ Hz} \tag{7.2.16}$$

This is more than 10 times lower than the excitation frequency. Thus we've got exactly the same situation that existed when we vibrated our hand at a much higher frequency than the system's natural frequency. End result? No motion of the big mass. Thus we can improve our approximation by saying that the sprung mass of the vehicle stays fixed, and we're left with the wheels being restrained by two sets of springs, namely, the suspension and the tire springs. Since these are acting in parallel on the sprung mass, we'll have

$$73\ddot{x}_h + 2(184{,}000 + 18{,}700)x_h = 0 \tag{7.2.17}$$

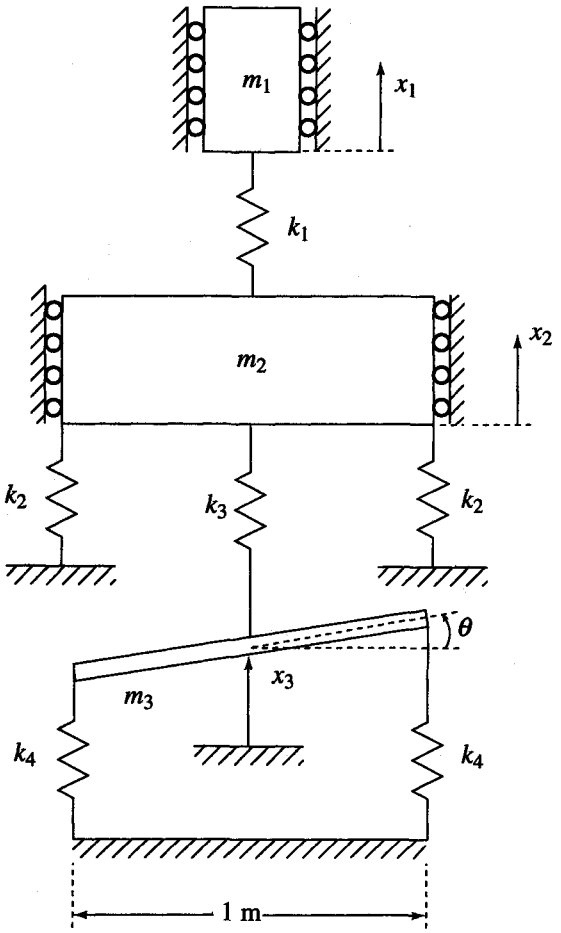
which gives us a natural frequency of

$$\omega_n = \sqrt{\frac{405{,}400}{73}} = 74.5 \text{ rad/s} = 11.86 \text{ Hz} \tag{7.2.18}$$

The result? An amazingly good approximation, one in which the error is less than a tenth of a percent.

Now that you've seen the power of this kind of analysis, let's review more carefully what we're doing and see when it will work and when not. First of all, the approach depended upon the existence of distinct and *identifiable* modes of vibration in our system. For a general system, there might be no easily identified modes of the sort we have in the car problem. The first mode might involve both front wheels moving up and down in unison while the rear ones moved antisymmetrically. Or it might have involved equal amounts of bounce and roll happening simultaneously. If this had been the case, our assumptions of pure bounce or pure wheel hop would have been wrong and the frequency estimates associated with them would have been equally wrong. This is where engineering intuition first comes in. Whether from analyzing different cars in the past or from an examination of the model, you have to suspect that the bounce, roll, and hop modes will show up. Only then can you solve for their frequencies of vibration.

After you've decided that certain physical modes should predominate in the response, you've still got to determine the essential factors that govern its motion. For the bounce mode, the key fact was in deciding that the two springs (tire and suspension) could be modeled as a series of springs, with a resultant equivalent single spring. For the wheel hop mode, we had to use a bit more insight and realize that certain elements of the vehicle wouldn't be involved in the motions (the sprung mass) because they were isolated from the high-frequency vibrations by their relatively soft springs. If the suspension had been a very hard one, one in which the suspension spring constants were as large as the tire spring constants, then the sprung mass *wouldn't* have been isolated and would have moved along with the wheels. This would have complicated our analysis.



$m_1 = 40$ kg
$m_2 = 1500$ kg
$m_3 = 48$ kg
$k_1 = 50{,}000$ N/m
$k_2 = 10{,}000$ N/m
$k_3 = 5000$ N/m
$k_4 = 100{,}000$ N/m
$I_3 = 4$ N/m

Figure 7.4 4 DOF system

Example 7.1

Problem To illustrate the concepts we've just learned, let's analyze the system shown in Figure 7.4. What we want to do is determine the modes of vibration and their natural frequencies without solving the entire coupled problem.

Solution As you can see, we have three masses and six springs. The first item of business is to determine how many modes there will be. m_1 can only move vertically, as can m_2. However, m_3 can both translate vertically and rotate. Thus this mass supports two degrees of freedom. This means that our system has a total of four degrees of freedom, and we expect to find four modes of oscillation.

First we'll note the relative masses: m_1 and m_3 are tiny in comparison to m_2. It seems very likely that for the lowest frequency mode, we'll have m_2 ponderously going up and down, dragging everything else along with it. To find this mode, we need to determine the equivalent spring that's restraining m_2. In total, we have two k_2 springs plus an equivalent spring that's made up of the k_3

spring in series with two k_4 springs. Thus the overall spring constant is

$$k_{\text{equiv}} = 2k_2 + \frac{k_3(2k_4)}{k_3 + 2k_4} = 24{,}878 \text{ N/m}$$

using the values given in Figure 7.4. The simple answer for the oscillations of this mass is to set it equal to 1500 + 40 which is the total of m_1 and m_2. The trickier move is to realize that m_3 is basically along for the ride, adding inertia but not doing much else in this low frequency mode. Since we know the spring constants surrounding this mass, we can determine how much it moves for a given motion of the main mass m_2. We'll just do a static analysis, like that shown in Figure 7.5. If we move the m_2 by x, then we need to do a force balance to decide how much m_3 moves. The force balance gives us

$$200{,}000y = 5000(x - y)$$

or

$$y = \frac{5000}{205{,}000}x$$

Thus the effect of m_3 will be to add $\left(\frac{5000}{205{,}000}\right)(48 \text{ kg}) = 1.17$ kg to the overall mass. This reflects the fact that m_3 doesn't move anywhere near as much as m_2, so we shouldn't include all its inertia.

Using our equivalent mass and spring, we can now solve for the natural frequency

$$\omega_n = \sqrt{\frac{24{,}878}{1541}} = 4.018 \text{ rad/s}$$

We can also find the approximate mode itself. If we use the coordinates

$$X^T = \{x_1 \;\; x_2 \;\; x_3 \;\; \theta\}$$

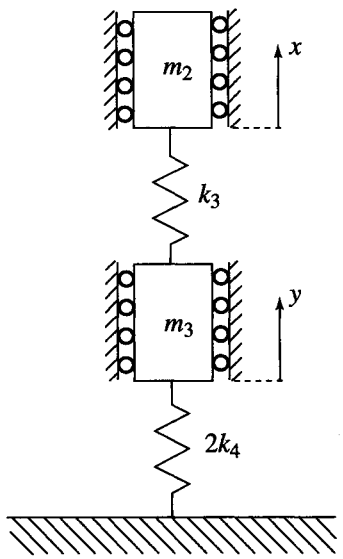


Figure 7.5 Static analysis to find equivalent mass contribution of m_3

(as shown in Figure 7.4), then we see that the mode is equal to

$$\tilde{X}_1^T = \{1.0 \;\; 1.0 \;\; .024 \;\; 0\}$$

where the tilde shows that this is just our estimate, not the actual eigenvector. This eigenvector was constructed by setting the response amplitude of x_2 to 1.0. Since m_1 is modeled as being rigidly attached to m_2, it also has amplitude 1.0. We just saw that the motion of m_3 is $\frac{5}{205}$ the size of m_2, which in this case is .024. Finally, the rotational mode isn't excited in this motion. Therefore it has an amplitude of zero.

Solving the fully coupled problem exactly results in a frequency of 4.019 rad/s and a mode shape of

$$X_1^T = \{1.000 \;\; .987 \;\; .024 \;\; 0\}$$

You can't get much better than this. There's hardly any difference at all between the approximate and exact results.

Now how about the other modes? One has already been alluded to. If m_3 is given an initial rotation, it will experience a rotary vibration. All the other motions in the system are purely translational, and there's no coupling between rotation and translation for this system. Thus we can analyze this mode immediately by setting all translations to zero and examining the rotational mode that's left. Note that this would definitely not be the case if the problem wasn't as symmetric as it is. For our system, m_3 is uniform, the springs from m_3 to ground are identical, and the attachments of the k_4 and k_3 springs are symmetric. These properties are what let us state that the rotations will not couple with the translational motion.

Taking the rotational part by itself, we have

$$4\ddot{\theta} + 2(100{,}000)\left(\frac{1}{2}\right)^2 \theta = 0$$

where the factor of 4 comes from the rotational inertia of m_3 and the rotational spring term comes from determining the restoring moment due to the k_4 springs. This supports a natural frequency of

$$\omega_n = 111.8 \text{ rad/s}$$

and an eigenmode estimate of

$$\tilde{X}_2^T = \{0 \;\; 0 \;\; 0 \;\; 1\}$$

Since there really *are* no other interactions for this mode, the exact solution and the approximate one are identical.

For our next approximate solution, let's examine the small mass m_1 that's attached to the very large mass m_2. Common sense tells us that the vibrations of a very small mass aren't likely to do much to affect the motion of a huge one. Thus we can approximate our next mode as one for which m_1 oscillates and m_2 remains fixed. This leaves us with the following SDOF problem:

$$40\ddot{x}_1 + 50{,}000x_1 = 0$$

The associated natural frequency is equal to 35.36 rad/s, and the approximate mode is given by

$$\tilde{X}_3^T = \{1 \;\; 0 \;\; 0 \;\; 0\}$$

That was certainly quick. Does it agree with the exact answer? Well, the exact solution will produce a natural frequency of 35.83 rad/s and an eigenvector of

$$X_3^T = \{1.000 \quad -.027 \quad -.001 \quad 0\}$$

You'd have to be a very stubborn individual not to agree that these two results are just about identical.

Finally, we have to address what's happening with the last mode. Calling on the same insight as that used in the car problem, we can check to see if there's a mode that oscillates at a much higher frequency than the large mass would want to do, thus isolating it. The first choice is to look at the motion that hasn't gotten any play yet, namely, translational motions of m_3. If we rearrange our system as shown in Figure 7.6, we can view it as two SDOF systems joined by a central spring. (We've neglected the m_1-k_1 combination because it's too small to worry about.) As shown in Figure 7.6, m_2 is sprung to ground through the two k_2 springs and attached to m_3 through k_3. We've already seen that the mode involving m_2 occurs at 4.018 rad/s. If we examine the m_3-k_3-k_4 combination, we've got a natural frequency of $\omega_n = \sqrt{\frac{205{,}000}{48}} = 65.35$ rad/s. Thus m_3 would like to oscillate at a much higher frequency than the low frequency mode of m_2. Therefore m_2 is isolated from the vibrations, and we can approximate it as being stationary.

If m_2 is taken as stationary, then we know from the preceding paragraph that the natural frequency associated with m_3's motion is 65.35 rad/s. The mode shape would be

$$\tilde{X}_4^T = \{0 \quad 0 \quad 1 \quad 0\}$$

The actual eigenvector is

$$X_4^T = \{.000 \quad -0.001 \quad 1.000 \quad .000\}$$

and the natural frequency occurs at 65.35 rad/s. Not much difference, eh?

It wasn't just an accident that the approximation results were so close to the exact ones; the system was designed so that this would happen. The point is that if you hadn't taken the time to think

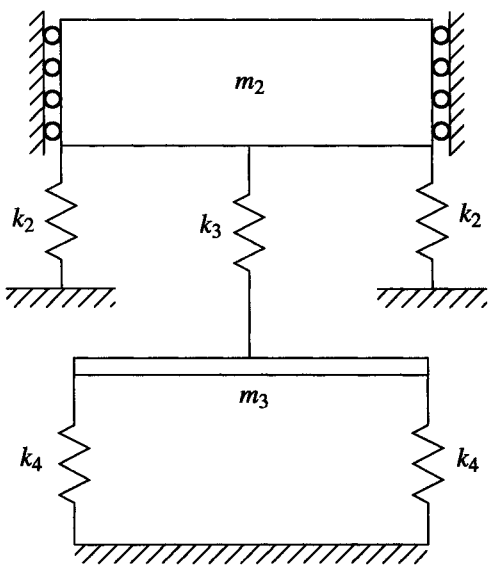


Figure 7.6 Reduced system ignoring m_1 and k_1

about what the system represented and considered ways to get approximate solutions, you'd have been forced to solve the full-blown, 4 DOF problem to get any results—and this isn't always the most intelligent approach. Real systems will invariably be complex, and solving them will require time and effort. If all you really want is an estimate of a couple of eigensolutions, it's worth your while to think a bit about intelligent approximations before jumping in.

7.3 LIMITING CASES

Up to now, our examples have had finite values for the mass and spring stiffness. Although this is very reasonable, it is also of use to consider what happens when one or both of these terms goes to zero or infinity. This is true for a variety of reasons. For one thing, we may wish to suggest design changes involving large reductions of mass or increases in spring stiffness in a given structure. In these cases, knowing the limiting value response for zero mass or infinite spring stiffness will let us know the greatest effect such an alteration could have on our system. In addition, an easy way to alter the system we're considering, if it's coded on the computer, is to have the computer make certain mass or stiffness elements very small or very large (approximating zero and infinity, respectively). This will serve to lock out or release degrees of freedom. Again, to correctly interpret the computer's output, we would want to know the analytical effect of this ahead of time.

This last technique, that of driving some terms to zero or infinity, is perhaps most useful in debugging programs. When the program is complex, it's hard to determine whether the output is really reasonable. You can often push the system into a simpler configuration, one that's easier to check, if you set certain parameters to their limiting values.

In Chapter 1 we considered the case of zero spring stiffness. Equation (1.2.2) showed the response of a free mass, which is the same as a mass-spring system for which the spring constant has been set to zero. As Equation (1.2.2) indicated, the mass can have an initial displacement and may also translate at a constant velocity if no forces are applied. This behavior is also indicated from our solution of a finite spring-mass system. Since the natural frequency of such a system is equal to the square root of $\frac{k}{m}$, we see that the consequence of k going to zero is that the natural frequency also goes to zero. And a zero frequency implies that the period of the oscillation is infinity. Thus the mass will move off and not come back. This is a very important point to remember: zero frequency implies translation (or rotation) with no oscillations. It is the free body motion referred to earlier. Note that increasing the mass a great deal while holding k fixed will give us the same result; it drives the natural frequency to zero.

This observation also has a more compelling intuitive feel. It is easy to see, given a huge mass and a tiny spring, that once the mass has started to move, the spring is going to have a very hard time reversing that motion. Since the force generated in the spring is directly proportional to its stretch, it's also clear that the little spring will have to stretch a great deal to generate sufficient force to bring the mass back; i.e., the mass will translate for a long time before the spring can affect it appreciably.

Next we'll look at the case of the mass going to zero (or equivalently, the stiffness going to infinity). In this case, the natural frequency will go to infinity, since either m going to zero or k going to infinity will cause $\frac{k}{m}$ to go to infinity. This makes sense if you consider a fixed mass attached to

a spring that gets stiffer and stiffer. As the stiffness becomes huge, the amount of motion needed to counteract any inertial forces is quite small. The system looks more and more like a rigid body. This is exactly the opposite of the preceding case. For small stiffnesses, the system became free. Now we see that large stiffnesses lock the system up.

These SDOF examples also hold when we look at MDOF systems. For instance, let's look at the system shown in Figure 7.7. This system consists of three masses and four springs. We'll fix the values of m_1-m_3 and k_1-k_3 but let k_4 vary from zero to 12 N/m. As we'll see, this range will show all the effects that we'd have had if we'd varied the stiffness from zero to infinity.

Since there are three masses, the system will display three distinct natural frequencies for each value of k_4. The variation of ω_i^2 is shown in Figure 7.8 for all k_4 values between zero and 12 N/m.

We already know that because we're adding stiffness to the system, we should expect all the natural frequencies to increase as k_4 is increased. We can also predict what the final frequencies

$$\begin{aligned} m_1 &= 1 \text{ kg} \\ m_2 &= 1 \text{ kg} \\ m_3 &= 1 \text{ kg} \\ k_1 &= 1 \text{ N/m} \\ k_2 &= 1 \text{ N/m} \\ k_3 &= 1 \text{ N/m} \\ k_4 &= 0\text{–}12 \text{ N/m} \end{aligned}$$

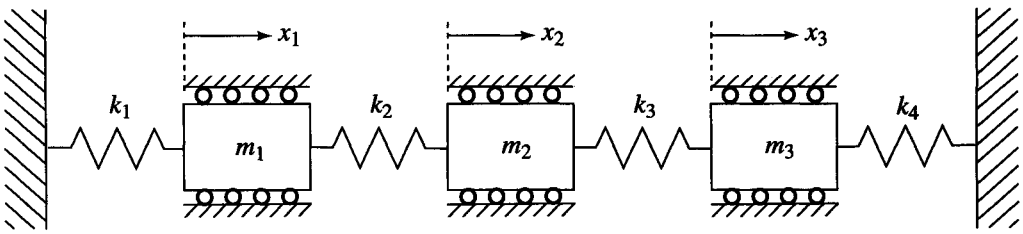


Figure 7.7 Spring-mass system with variable end spring

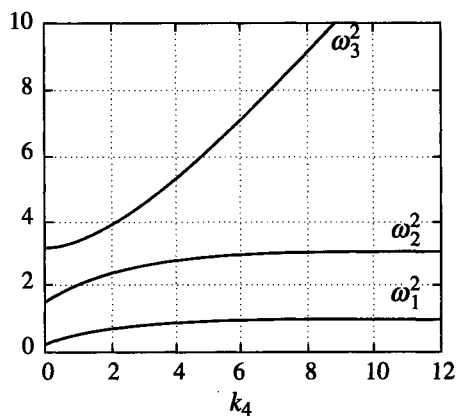


Figure 7.8 Variation of natural frequencies as a function of spring constant k_4

will be. When k_4 becomes huge, it locks down the motions of m_3. Thus we're left with a 2 DOF system. Analytically, we can solve for the natural frequencies, and find that they are 1 and $\sqrt{3}$ rad/s. Since Figure 7.8 plots the squares of the natural frequencies (the eigenvalues of the problem), we should see results of 1 and 3 for large k_4, which we do. In fact, you'll notice that essentially all the variation of the first two natural frequencies has occurred by the time k_4 has reached 8 N/m. This is quite typical. As soon as the effect of a particular element has become sufficiently large, the effect on the overall system response is *as if* it were equal to infinity.

We can also see that the third natural frequency does not level off but in fact climbs quite steeply as k_4 is increased. This frequency is associated with the reduced system of m_3, k_3, and k_4. For high values of k_4, the other masses are unaffected; the vibration is too high for them to respond. Thus m_1 and m_2 are essentially stationary, leaving the mass m_3 to vibrate, restrained by k_3 and k_4. The natural frequency squared of this mode is given by

$$\omega_3^2 = \frac{1 + k_4}{1} \tag{7.3.1}$$

(i.e., $\frac{k_3+k_4}{m_3}$ with the numerical values of m_3 and k_3 equal to 1). Therefore we would expect it to rise linearly as soon as k_4 becomes substantially greater than 1, which it does.

Next, let's consider small values of k_4. Setting k_4 at zero still leaves us with a 3 DOF problem. If you go ahead and solve for the exact natural frequencies for this case, you'll see that they're equal to those given in Figure 7.8 for the case of $k_4 = 0$.

You also will obtain similar results by letting damping and/or mass elements vary from small values to very high values. For instance, letting a mass element go to infinity will mean that you'll have a natural frequency that goes to zero (because of the huge mass) while the others go to the values they'd have if the mass were fixed in space. Damping can also lock out degrees of freedom. When the damping goes to infinity, it's as if the mass is moving in incredibly thick tar. Obviously, it won't move much in this case.

These insights can and should be used in your engineering work. Let's say you want to increase a natural frequency in a system because one of the current natural frequencies is near an external excitation frequency. You may want to move it up by 20% and you may think you can do this by increasing a certain spring stiffness in the system. By analytically setting this stiffness to infinity, and analyzing the resulting vibrations, you can determine what the maximum frequency changes are going to be. If they're not sufficient, you know not to waste time trying to alter that particular spring—it's not going to be sufficient to do the job.

7.4 VERIFYING YOUR ANALYSES

One of the most important topics that you'll need to master is verifying your analyses. You may have developed a nice analytical or approximate model for a system and you're ready to produce some results. But how do you know if your model is any good? Or perhaps you've included some typos in

your computer code, typos that don't keep the code from running but do give you incorrect answers. How can you tell?

The first secret is a simple one. Go slow. Go REALLY SLOOOOWWWW. This probably sounds silly, but it isn't. It's *incredibly* easy to make an error when flying through a derivation. You should try to develop the discipline to go a line at a time and *check each line*. You'll catch more than enough errors to justify the time spent in checking.

Okay, let's say you went slowly and did what you think is a good job. What now? Well, you can check that the limiting cases of your written equations work out. For instance, do your equations give you the correct static result? If you go through your equations and cross off all the time-dependent terms, you'll be left with a static equation, one that's easy to solve. It's usually pretty straightforward to determine what the physical response should be for a static situation. Thus you can check that at least parts of your equations are correct. The same reasoning also applies to the other extreme. If you kill all but the highest frequency terms, you'll be left with equations that describe the upper frequency limit for your problem, something you can then verify.

If you're attempting a modal approximation, you could include only a single mode to start. If your system is restrained to ground, this could be a rigid body mode. Your approximate analysis should then match the results you'd get from letting the distributed system behave rigidly. If it doesn't, you've got an error somewhere.

Most of the time, you'll end up with computer code that's supposed to accurately represent your equations, which in turn represents some physical reality. By looking at limiting cases for all of these, and cross-checking, you'll go a long way toward verifying that your work is correct.

7.5 HOMEWORK PROBLEMS

Section 7.2

7.1. Figure P7.1 shows two snapshots of a tensioned string undergoing free vibration. Explain how such a situation could arise if the system modes were proportionally damped. Explain how it could also occur under nonproportional damping.

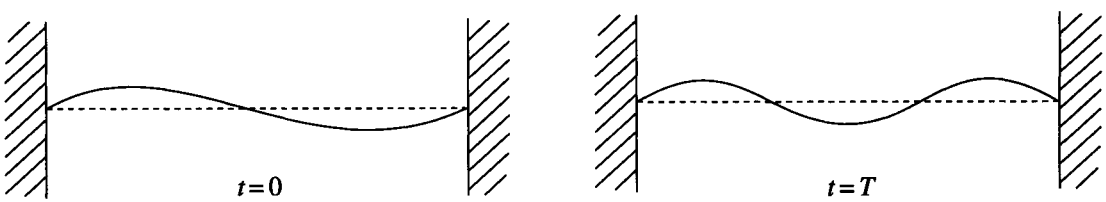


Figure P7.1

7.2. Your research assistant has told you that three given vectors represent the three eigenvectors of a particular system. You put these together into the model matrix $[U]$ and form the matrices $[U]^T[M][U]$ and

$[U]^T[K][U]$, which turn out to be

$$[U]^T[M][U] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad [U]^T[K][U] = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

What can you say about the three vectors?

7.3. Figure P7.3 shows a seismically excited system. For an input displacement $y = \bar{y}(\cos \omega t)$, what will the phase lag of the output x to input y equal? An answer that's within 10% is good enough. $m = 22$ kg, $k = 20{,}000$ N/m, $c = 135$ N·s/m, $\bar{y} = .1$ cm, $\omega = 600$ rad/s.

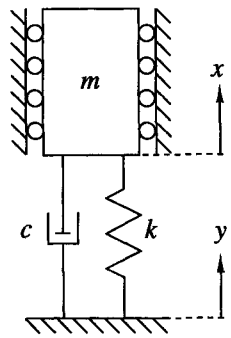


Figure P7.3

7.4. Consider the system shown in Figure P7.4: it's made up of a torsional rod with a heavy mass having a rotational inertia J_0 at one end. An external moment M is applied at $x = l$. For what frequency range (in terms of ρ, G, and J) would it be appropriate to model this system as an SDOF system (i.e., the lumped inertia of the mass at the end and the rod stiffness act together as a SDOF spring-mass system)?

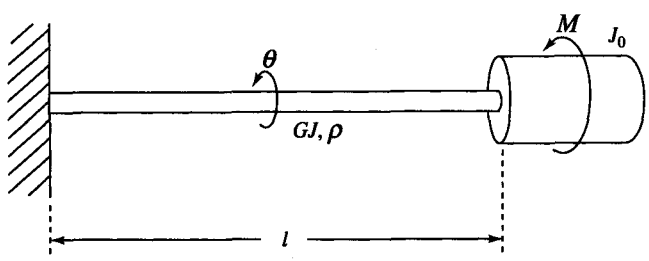


Figure P7.4

7.5. What response variable would produce the amplitude/phase outputs shown in Figure P7.5? (*Hint*: Consider the types of system we've seen—(mass excited, rotating imbalance, seismically excited—and what could

be measured—$x, \dot{x}, \ddot{x}$, etc.). Note that the magnitude of the output goes to zero as the forcing frequency goes to infinity.

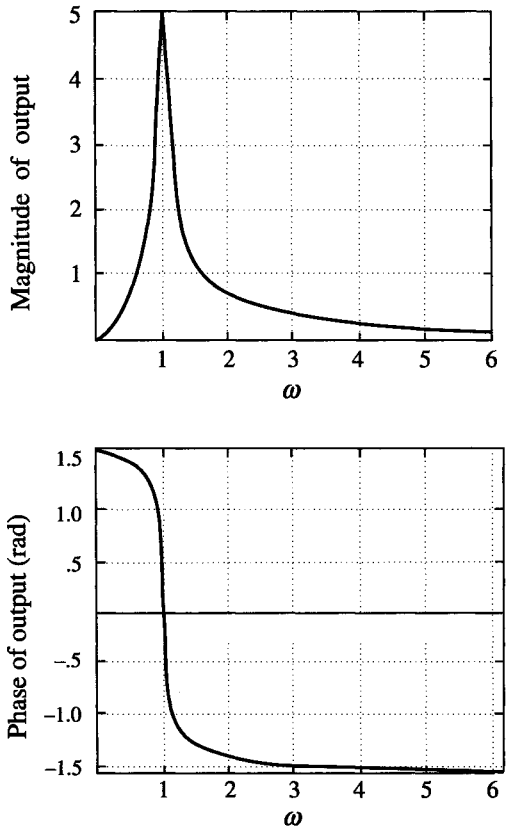


Figure P7.5

7.6. Consider the tensioned string with a string-mass system suspended from it, as shown in Figure P7.6. The string can be forced at a variety of frequencies. If the forcing is equal to the frequency of the first string mode (ignoring the spring-mass), what should k/m be to minimize the response? What if the forcing was at the second natural frequency? Can the spring-mass system be placed anywhere along the string in both these cases, or are there restrictions on its location? Note that this is NOT a Rayleigh-Ritz problem. $\rho = 1$ kg/m, $l = 1$ m, $T = 1$ N.

7.7. Find the eigenvectors and natural frequencies for the system illustrated in Figure P.7.7. Describe the modes that seem to involve only part of the structure and explain why the whole structure (all x_i's) isn't involved. $m_1 = 1$ kg, $m_2 = 10$ kg, $m_3 = 50$ kg, $m_4 = 1$ kg, $m_5 = 200$ kg, $k_0 = 1000$ N/m, $k_1 = 10$ N/m, $k_2 = 50$ N/m, $k_3 = 75$ N/m, $k_4 = 2000$ N/m, $k_5 = 1$ N/m, $k_6 = 2000$ N/m.

7.8. What's wrong with the amplitude-versus-frequency response illustrated in Figure P7.8?

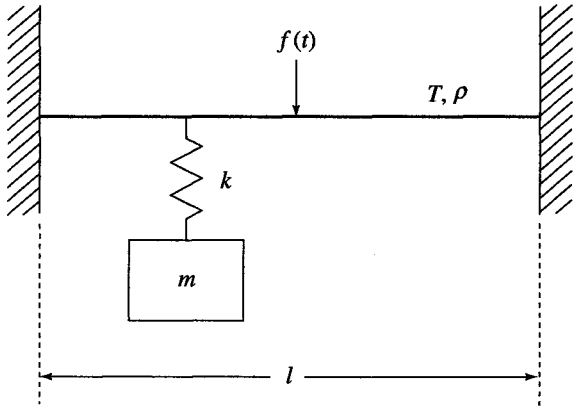


Figure P7.6

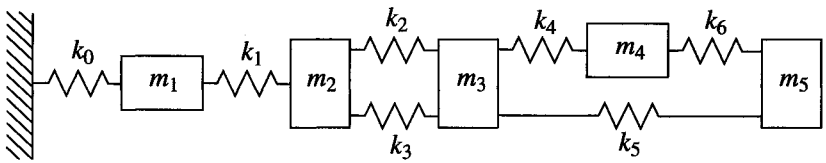


Figure P7.7

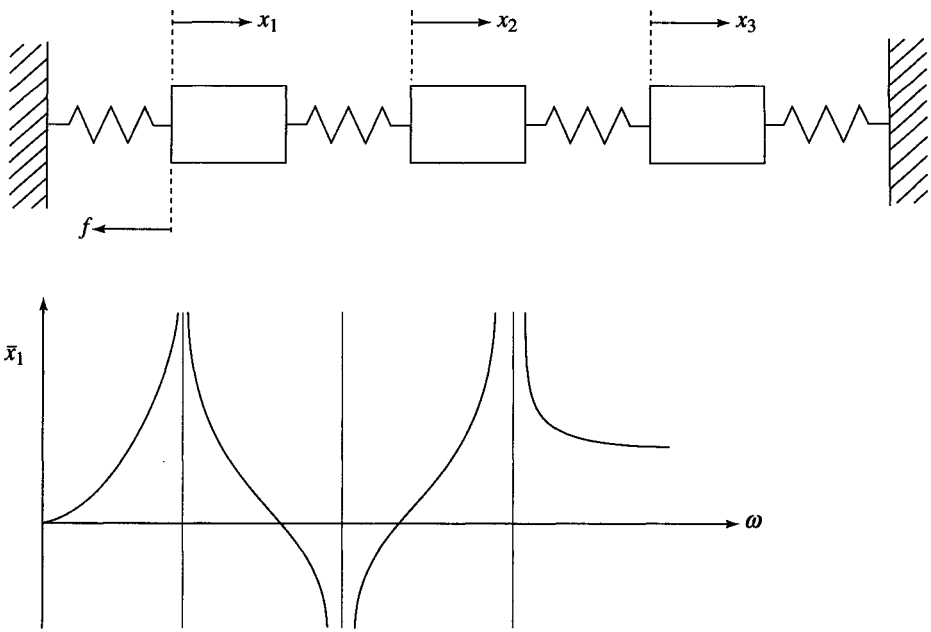


Figure P7.8

7.9. Show how you would find the *approximate* natural frequencies and eigenvectors for the system illustrated in Figure P7.9. Don't solve it exactly. The interest here is in how you'd approach it in an approximate sense. Then solve it exactly and compare the results. $k_i = 10$ N/m, $m_1 = 1$ kg, $m_2 = 100$ kg, $m_3 = 2$ kg.

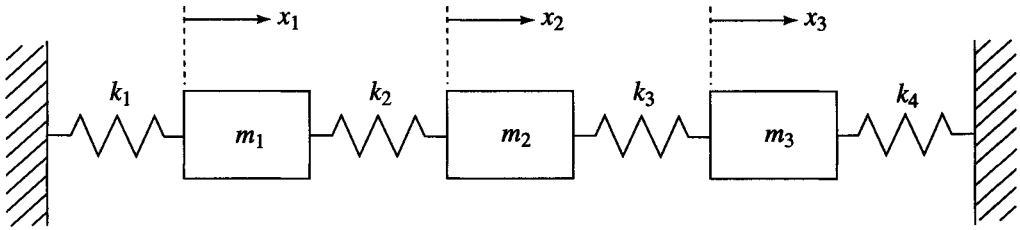


Figure P7.9

7.10. In this problem we'll try to gain some insight with respect the analyses of simpler subsystems, to determine when is a valid approach. We'll consider the four different configurations of the relative masses and stiffnesses of Figure P7.10. Solve the problems exactly and comment on which aspects of the solution are predictable from an analysis of a simpler subsystem. For all the problems, $f_i = 0$ and $k_3 = 0$, the base system is shown in part (a).

$$(a)\quad \begin{array}{ll} m_1 = 100 \text{ kg} & \\ m_2 = 1 \text{ kg} & \text{base system} \\ k_1 = 1000 \text{ N/m} & \\ k_2 = 10 \text{ N/m} & \end{array}$$

$$(b)\quad \begin{array}{ll} m_1 = 100 \text{ kg} & k_1 = 1000 \text{ N/m} \\ m_2 = 1 \text{ kg} & k_2 = 1000 \text{ N/m} \end{array}$$

$$(c)\quad \begin{array}{ll} m_1 = 100 \text{ kg} & k_1 = 1000 \text{ N/m} \\ m_2 = 100 \text{ kg} & k_2 = 1000 \text{ N/m} \end{array}$$

$$(d)\quad \begin{array}{ll} m_1 = 100 \text{ kg} & k_1 = 1000 \text{ N/m} \\ m_2 = 100 \text{ kg} & k_2 = 10 \text{ N/m} \end{array}$$

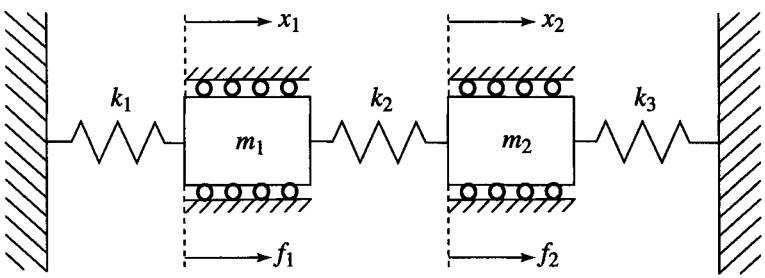
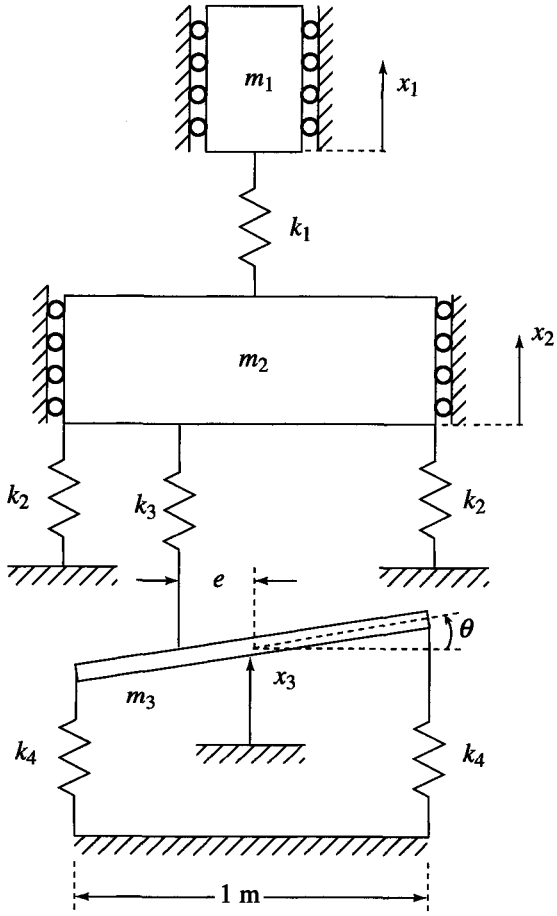


Figure P7.10

7.11. Let's spend some more time with Example 7.1 (Section 7.2). A revised version of the system is shown in Figure P7.11. The change is the fact that k_3 is no longer centered, but is offset by an amount e. Determine how much effect such a misalignment can have. Do the changes in the frequencies and eigenvectors make physical sense?



$m_1 = 40$ kg
$m_2 = 1500$ kg
$m_3 = 48$ kg
$k_1 = 50{,}000$ N/m
$k_2 = 10{,}000$ N/m
$k_3 = 5000$ N/m
$k_4 = 100{,}000$ N/m
$I_3 = 4$ N/m

Figure P7.11

7.12. In this problem we'll again consider Example 7.1. This time, you should vary the mass m_1. When it was small, it decoupled nicely from the rest of the system. As its mass increases, the assumption of decoupling will become invalid. At what mass do the errors in the frequency estimates of Example 7.1 exceed 10% in comparision to exact results?

7.13. The system illustrated in Figure P7.13 is a horizontal beam supported by two vertical beams. We want to determine the appropriate boundary conditions to use with the horizontal beam. One possibility is to view the ends as fixed. But perhaps this isn't reasonable. After all, we know that bars support longitudinal

motions and the vertical vibrations of the horizontal beam might interact with the vertical compression modes of the two supporting beams. In this last case the supporting beams are actually acting like bars in longitudinal motion.

Look at the eigenfrequencies and deduce which boundary condition makes the most sense—fixed or spring restrained (the vertical supports providing the spring forces). All elements of the structure have the same parameter values. $E = 2.1 \times 10^{11}$ N/m, $\rho = 7.8 \times 10^3$ kg/m^3, $l = 1$ m, $A = .0001$ m^2, $\bar{I} = 8.\bar{3} \times 10^{-10}$ m^4.

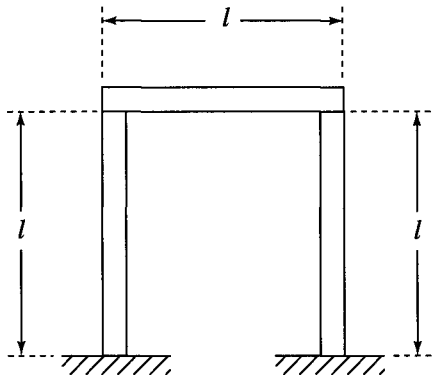


Figure P7.13

Section 7.3

7.14. The system illustrated in Figure P7.14 is composed of a lumped mass m_1 that's connected to ground by means of two tensioned strings. The length of each string is 1 m. The linear density of the strings is 2 kg/m, the lumped mass is variable, and the tension is equal to 1800 N. A lumped approximation (ignoring the string's mass) produces a natural frequency of $\sqrt{\frac{2T}{m_1}}$. How small can m_1 be before the actual first natural frequency of the string/mass system differs from this estimate by 10%?

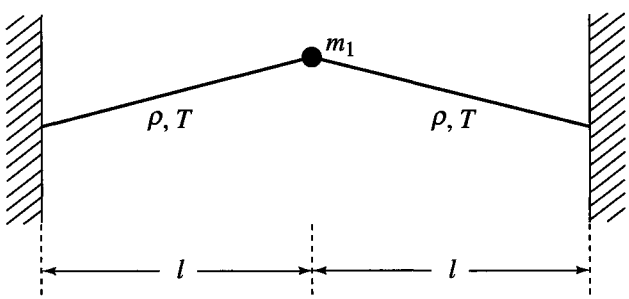


Figure P7.14

7.15. Figure P7.15 shows, a uniform, rigid bar, spring-restrained to ground, with a lumped mass attached to it. The lumped mass can be placed anywhere along the bar. Analyze the system for different placements of

the mass (from midspan to the end) and comment on how badly the symmetry of the solution (when the mass is at midspan) is affected by its movement off from the center. $l = 2$ m, $m_2 = 3$ kg, $m_1 = 1$ kg, $k = 10{,}000$ N/m.

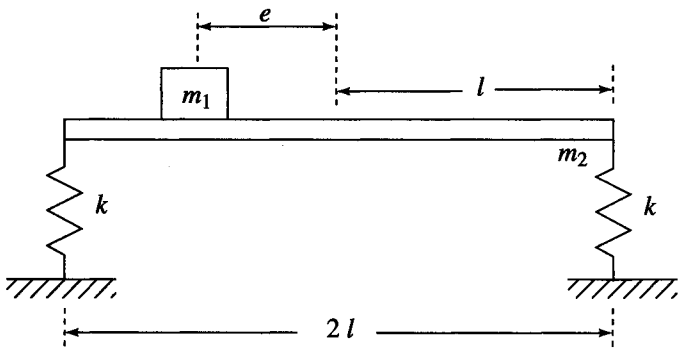


Figure P7.15

7.16. Figure P7.16 shows a uniform, rigid bar that's spring-restrained to ground. Analyze the system for the case of equal spring constants and then allow the right spring to become progressively stiffer. How do the solutions alter as the spring constant is increased? How much can it change and still leave the symmetric solutions (associated with equal spring constants) relatively unchanged? $m_1 = 1$ kg, $k_1 = 1000$ N/m, $l = 1$ m.

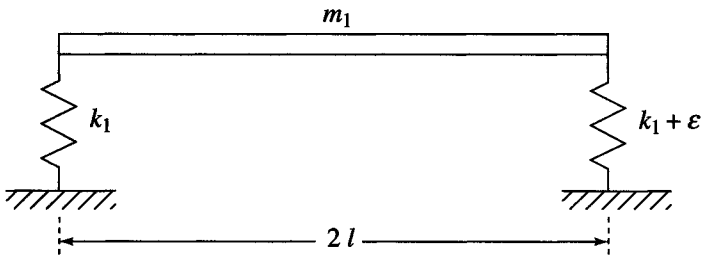


Figure P7.16

7.17. Figure P7.17 shows a tensioned string with an attached mass ε. All motion takes place in a horizontal plane, and the mass moves without friction along the guide. If the mass is large, the first vibratory mode can be analyzed as if the system is SDOF, with the string tension supplying the restoring force and the lumped mass acting as the dominant mass in the system. As the mass reduced, however, the other modes of the string problem become more important and, eventually, the SDOF approximation cases to be accurate. At what value of ε does the SDOF approximation for the lowest natural frequency exceed 5% error compared with the actual solution of the coupled string-mass problem? The tension in the string is 100,000 N, $\rho = 1$ kg/m, $l = 1$ m. Neglect gravity.

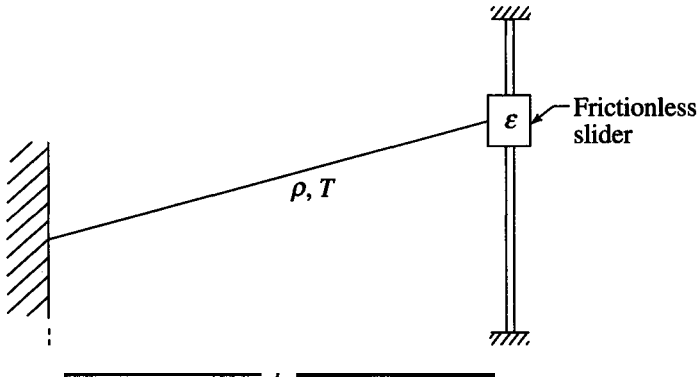


Figure P7.17

7.18. Consider again the 2 DOF system shown in Figure 7.10. Let $f_1 = f_2 = 0$ and let $k_3 = 0$. If k_1 is reduced sufficiently, the system can be viewed as supporting two decoupled modes. One involves motion of both m_1 and m_2, which move like a rigid body. The other involves the two masses moving in opposition to each other, with k_1 contributing little to the response (a *breathing* mode). Both these modes are shown in Figure P7.18. For what values of k_1 is the assumption of modes like those shown in the figure a good one? (*Note*: You'll have to define an error measure for when the modes look reasonable.) $m_1 = m_2 = 1$ kg, and $k_2 = 900$ N/m.

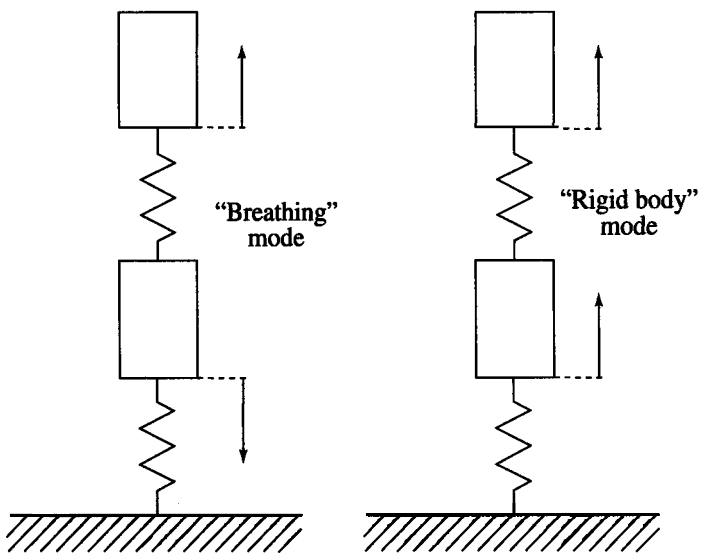


Figure P7.18

7.19. Consider a tensioned string with a lumped mass and spring at the right end, as shown in Figure P7.19. Solve for the actual natural frequencies. Next, use your insight into how the first mode will differ qualitatively

from the higher ones and show how you can obtain an approximate first natural frequency without solving the entire continuous problem. Don't forget to include the equivalent lumped mass and stiffness provided by the string. $l = 1$ m, $m_1 = 10$ kg, $k = 9000$ N/m, $\rho = .1$ kg/m, $T = 50$ N.

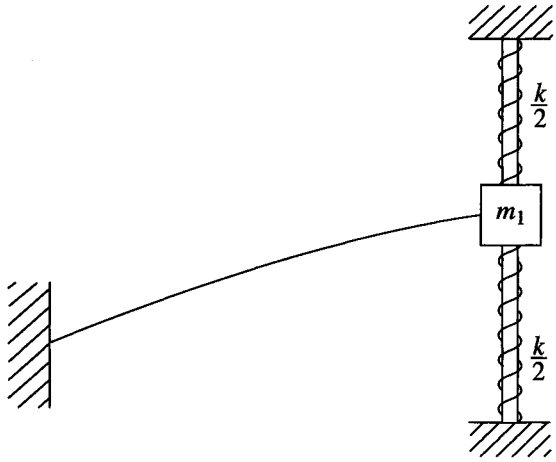


Figure P7.19

7.20. Assume that you've solved for the responses of the undamped vibration absorber problem shown in Figure 7.20. $m_1 = 1000$ kg, $k_1 = 36,000$ N/m, $m_2 = 2$ kg, $k_2 = 72$ N/m, $\omega = 6.03$ rad/s, and the forcing amplitude is .7 N. You determine that $x_1 = -.00783$ m and $x_2 = .781$ m. Verify that these results make sense by analyzing the SDOF system made up of m_2 and k_2, presuming that the system is seismically excited by m_1.

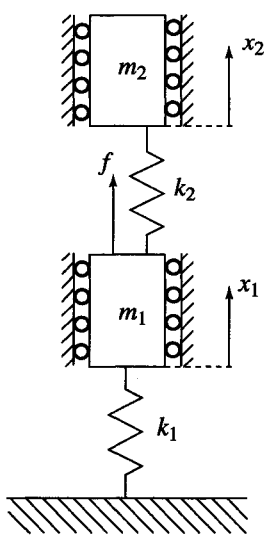


Figure P7.20